Different Points of Departure in Statistical Physics, and the Paths They Indicate

— A Discussion of Particle Ontology and Structure Ontology

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Preliminary Note

In the study of physical problems, one typically adopts a particular theoretical point of departure. In statistical physics, the present work classifies such points of departure into two types: particle ontology and structure ontology. Different choices of theoretical starting point lead to different paths of interpretation, which are reflected in differing descriptions of the same physical phenomena. The discussion that follows is organized along this line. At the same time, this work also considers the significance of structure ontology. Its significance lies in the possibility of placing several seemingly distinct thermodynamic phenomena within a unified statistical framework.

1 The Origins of the Two Points of Departure

The origin of particle ontology. Early studies in physics proceeded from entities as their point of departure. For example, in mechanics, the motion of objects and the interactions between them are investigated. In thermodynamics, matter is regarded as composed of a large number of microscopic particles, and macroscopic thermodynamic phenomena are described on this basis. In atomic physics, the internal structure of atoms serves as the primary object of study, focusing on the motion of particles within atoms and their interactions. This includes, in statistical physics, the Boltzmann distribution, which is also formulated on the basis of a particle model. Theories that take particles as their point of departure are here referred to as particle ontology.

The origins of structure ontology. At present, two lines of reasoning can be identified that point toward taking overall structure as the primary object of study. One line arises from the uncertainty relations in quantum mechanics. In particular, the expressions for position uncertainty and momentum uncertainty imply, in a formal sense, that the product of the uncertainties in position and momentum at a given moment has a lower bound. When this bound is approached, the particle becomes completely blurred, and the concreteness of the particle appears to lose its significance. This aspect suggests that, when considering certain problems, the object of study may need to be shifted, rather than taking particles as the primary focus. Another line arises from Bose's method of statistical counting, through which the blackbody radiation law is derived [1]. In this

method, phase space is divided into cells of volume h^3, emphasizing the occupation of quanta within these cells. Here, the cells function as the basic units for describing structure in phase space. These two lines of reasoning indicate that it is possible to take structure as the primary object of study. Theories that take structure as their point of departure are here referred to as structure ontology.

2 Different Paths Indicated by the Two Points of Departure

In statistical physics, the Boltzmann distribution and the Bose distribution may be regarded as corresponding to two distinct paths arising from different points of departure. The former takes particle as its starting point, whereas the latter takes structure as its starting point. Owing to the difference in points of departure, the aspects of the problem on which they focus are also different. Particle ontology emphasizes dynamical issues such as the motion and interaction of particles. By contrast, structure ontology focuses on the distribution of system energy among phase cells(in this framework, only the conservation of total energy in an equilibrium system is taken as an independent constraint, correspondingly, only one Lagrange multiplier is introduced). For this reason, the two approaches differ markedly in their understanding of equilibrium state, temperature, and the problem of system heating.

First, they understand equilibrium state differently. In particle ontology, equilibrium state is often understood as a state gradually formed through the interaction of a large number of particles and through the traversal of various possible paths; therefore, it is more easily interpreted as a state reached through temporal evolution [2]. By comparison, the equilibrium state understood by structure ontology does not involve descriptions of temporal processes, interactions, or causal chains. Rather, it understands equilibrium state as the stable statistical distribution of energy among phase cells under the constraint of conserved total energy.

Second, they interpret the concept of temperature differently. Particle ontology holds that temperature reflects the motion state of microscopic particles, the higher the temperature of the system, the more intense the motion of the microscopic particles; the lower the temperature of the system, the less intense the motion of the microscopic particles. Structure ontology, by contrast, holds that temperature characterizes the distribution state of energy over phase cells [3], the greater the average number of quanta within each phase cell, the higher the system temperature appears; the smaller the average number of quanta within each phase cell, the lower the system temperature appears. In this sense, temperature is understood as the statistical representation of distribution of system energy over phase cells.

Third, the two approaches give different accounts of the phenomenon of system heating. From the perspective of particle ontology, when external energy is transferred into the system, particles move from lower energy levels to higher energy levels, in this process, the energy of particles increases. From the perspective of structure ontology, however, this process may be described as follows: external energy is transferred into the system, and the system energy is redistributed among phase cells, this distribution follows the principle of maximum entropy(i.e.the most probable macrostate corresponds to the largest number of microscopic configurations). As a result, the average number of quanta in each phase cell increases, which appears macroscopically as a rise in system temperature. [3]

3 The Relation Between the Two Points of Departure

What needs to be clarified here is that, in statistical physics, the relation between particle ontology and structure ontology is not one of mutual negation, nor is there a simple distinction between right and wrong. Rather, the difference lies in the starting point chosen when studying a problem. Different starting points lead to different emphases on the same problem, different methods of investigation, and, consequently, different conclusions.

Particle ontology takes particles as fundamental, focusing on their states of motion and interactions. When studying a problem, it often examines how, after a certain period of time, particles change from one state to another, and how the interactions between particles operate. This easily leads toward the path of dynamics. Therefore, according to the interpretation of particle ontology, equilibrium state is a state ultimately reached through a dynamical process of particles; temperature is also understood as a macroscopic manifestation of the intensity of microscopic particle motion; and the phenomenon of system heating is interpreted as an increase in the energy of the particles themselves.

Structure ontology is different. It does not first ask whether particles exist as entities, but instead considers how the energy of the system is distributed over phase cells, which are the smallest units constituting the structure. In statistical physics, structure ontology takes phase cells as fundamental. In this way, equilibrium state, temperature, and the problem of system heating are understood as different forms of expression of the distribution of energy over phase cells.

In summary, particle ontology focuses on particles–their dynamical processes, whereas structure ontology focuses on phase cells–the mode in which energy is distributed. Below, this paper discusses the significance of structure ontology. In some cases, structure ontology offers a more concise and direct interpretation of certain problems and can place these problems within a unified statistical framework.

4 The Significance of Structure Ontology

The uncertainty relation in quantum mechanics indirectly indicates that, when studying certain problems, the object of study should be shifted from particles to other objects. Bose's derivation of blackbody radiation law by means of statistical methods further provided a new object of study: phase cells, which serve as the most basic units for describing structure. Thus, the door was opened to taking structure as the starting point.

In statistical mechanics, choosing structure ontology to describe problems can lead to concise and direct explanations in some cases. As noted above, structure ontology understands equilibrium state, temperature, and the phenomenon of system heating as different forms of expression of the distribution of energy over phase cells. This means that three seemingly different phenomena are placed within the same statistical framework: equilibrium state corresponds to a stable statistical distribution; temperature is a statistical representation of this distribution; and the phenomenon of system heating corresponds to a change in the statistical distribution.

Therefore, in statistical physics, structure ontology not only provides a new starting point for research questions and points toward an explanatory path different from that of particle ontology, but also provides a more concise and direct statistical description

of thermodynamic phenomena and places these phenomena within a unified statistical framework.

References

[1] Satyendra Nath Bose. Planck's law and the hypothesis of light quanta. *Z. Phys*, 26(178):1–5, 1924.

[2] Frederick Reif. *Fundamentals of statistical and thermal physics*. Waveland Press, 2009.

[3] Bo Yang. A discussion of phase cells as the basic descriptive units of structure in statitical mechanics. 2025. DOI:10.5281/zenodo.18468146.